

# EventTrace: Verifying Ordered Semantic Events for Reinforcement Fine-Tuning of Vision-Language Navigation

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## Abstract

001 Reinforcement fine-tuning of vision-language navigation  
002 (VLN) policies commonly relies on destination suc-  
003 cess or similarity to a reference route. These signals  
004 leave an instruction’s intermediate requirements unre-  
005 solved: reaching the endpoint does not establish that  
006 the agent entered the requested room or passed the  
007 specified landmark. We study an online reward that  
008 represents an instruction as an ordered list of seman-  
009 tic events and verifies each event against observations  
010 before and after executed actions. A frozen vision-  
011 language model supplies a three-way verdict (complete,  
012 incomplete, uncertain); motion and stopping checks re-  
013 ject visually plausible but unexecuted events. Each  
014 event is rewarded at most once, and the process re-  
015 ward is bounded below the success bonus. We inte-  
016 grate EventTrace into multi-turn ActiveVLN/GRPO  
017 and compare it with destination-only training for 64  
018 optimization steps across three matched seeds. All six  
019 checkpoints and the SFT initialization are evaluated  
020 on the same 1,839 R2R val-unseen episodes. Event-  
021 Trace changes success rate by +0.65, +0.44, and -2.88  
022 percentage points across seeds; its mean paired change  
023 is -0.60 points (sample standard deviation 1.98). A  
024 single-AI blind review of 49 selected transitions, re-  
025 played with training-time image encoding, finds that 7  
026 of 15 verifier completion predictions match its labels.  
027 Two subsequent independent human reviewers agree on  
028 only 33 of these 49 cases; their 16 disagreements remain  
029 unadjudicated. The method is operational, but these  
030 results do not establish a navigation improvement or  
031 reliable semantic verification.

## 032 1. Introduction

033 Vision-language navigation asks an embodied agent to  
034 follow a natural-language route through visual obser-  
035 vations. The Room-to-Room (R2R) task introduced

instruction-guided navigation in real indoor environ-  
ments [2]; VLN-CE moves this task into continuous  
control [7]. Success rate (SR) and success weighted by  
path length (SPL) capture destination attainment, yet  
a successful stop can coexist with skipped intermedi-  
ate instructions. Instruction fidelity work made this  
gap explicit [6].

Recent reinforcement fine-tuning (RFT) approaches  
give visual-language policies online or trajectory-level  
feedback. ActiveVLN performs multi-turn exploration  
with group-relative policy optimization (GRPO) [11,  
14]. VLN-R1 uses time-decayed action rewards [10];  
Nav-R1 combines format, understanding, and naviga-  
tion rewards [9]; SeeNav-Agent proposes step-level ver-  
ifiable process rewards in another navigation bench-  
mark [13]. Progress-Think learns semantic progress  
from visual histories and instruction prefixes [12];  
SACA audits stepwise landmark progress and routes  
mixed-success and all-failure groups differently [8].  
Our question is narrower: can a navigation instruc-  
tion’s ordered, physical events provide a bounded re-  
ward during closed-loop R2R-CE rollouts, without  
matching the policy’s actions to a reference path?

We introduce EventTrace, a modular reward on top  
of an existing multi-turn policy. The reward uses a  
short event list, a frozen visual verifier with an absten-  
tion state, deterministic checks of motion or stopping,  
and a once-only order-preserving tracker. The actor  
sees only the original egocentric observations and in-  
struction; simulator displacement is available only to  
the reward. Figure 1 illustrates the distinction between  
seeing a bathroom and actually entering it.

We evaluate the reward with matched three-seed  
training and complete R2R val-unseen coverage. The  
observed seed variation and an exploratory verifier au-  
dit limit the empirical claim: this implementation is  
reproducible but does not improve navigation consis-  
tently.

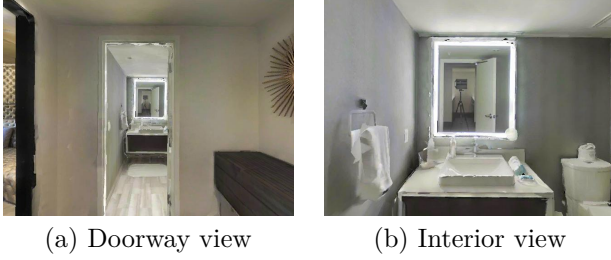


Figure 1. Real simulator frames from a diagnostic replay of R2R training episode 2889. Visibility alone does not complete the event “enter bathroom.” The verifier uses before/after views and executed motion; the ordered tracker later checks “stop near sink.” These frames illustrate the mechanism and are not an independent accuracy test.

## 2. Related Work

Path and instruction fidelity. SR and SPL reward the destination and efficiency, whereas nDTW and coverage weighted by length score (CLS) compare trajectories with a reference route [5, 6]. Reference trajectories can be useful supervision but do not directly answer whether a distinct visual event actually occurred when multiple valid routes exist. Our reward targets event completion during execution; route metrics remain complementary evaluation measures.

RL for VLN. ActiveVLN is our implementation base because its environment can return feedback after each multi-action turn [14]. VLN-R1 and Nav-R1 show other ways to shape VLN rewards [9, 10]. Progress-Think aligns observation and instruction progress and co-trains a progress-aware policy [12]; SACA combines a landmark auditor with scenario-dependent group optimization [8]. SeeNav-Agent emphasizes step-level verification [13]. EventTrace tests a narrower combination of a once-only ordered event state, an explicit uncertainty outcome, and motion gates against rewarding a landmark merely seen without the requested transition. Our pilot applies GRPO to trajectory-level returns; it does not claim turn-level credit assignment or stronger navigation performance than these systems.

## 3. Method

### 3.1. Task and event representation

An instruction  $x$  yields a list  $E(x) = (e_1, \dots, e_K)$ , where  $e_k$  contains a verb type, target phrase, and source span. Supported event types include enter, pass, descend, and stop\_near. A frozen vision-language model (VLM) proposes at most three events, restricted to source spans in the instruction. The eight-episode

Table 1. Motion gates in the pilot. The VLM must also verify that the image pair supports the specified target.

| Event     | Count | Required gate                              |
|-----------|-------|--------------------------------------------|
| enter     | 3     | Forward action; $\ \Delta p\  \geq 0.10$ m |
| pass      | 3     | Forward action; $\ \Delta p\  \geq 0.10$ m |
| descend   | 3     | Vertical change $\Delta y \leq -0.08$ m    |
| stop_near | 8     | Executed STOP action                       |

wiring pilot uses curated lists; the main 64-step training runs parse instructions automatically at episode start. Parser errors produce an empty list and no semantic reward. We have not independently audited parser accuracy.

In the curated pilot, each of the eight episodes contains a stopping event; there are also three enter, three pass, and three descend events, for 17 in total. These counts describe the pilot distribution, not the prevalence of event types in R2R. Table 1 shows the deterministic portion of the verifier. Each gate can reject completion before a VLM call, but passing a gate is never sufficient to award a point.

At turn  $t$ , the policy receives the instruction, action history, and current egocentric image  $I_t$ , and emits up to three low-level actions. The simulator executes these actions and returns  $I_{t+1}$ , pose change  $\Delta p_t$ , and task metrics. Only the current pending event  $e_{k_t}$  is queried. The frozen verifier produces

$$v_t = V(e_{k_t}, I_t, I_{t+1}, a_t, \Delta p_t) \in \{Y, N, U\}, \quad (1)$$

where U denotes insufficient evidence. A view of a landmark is not by itself a completion verdict. For crossing and passing, a forward displacement check is required; descending requires a negative vertical displacement; stopping near a target requires an executed STOP. The VLM inspects whether the image pair supports the specified target and transition. These are deliberately conservative gates, not a learned proof of physical state.

### 3.2. Ordered, bounded process reward

Let  $k_t \in \{1, \dots, K+1\}$  index the first unfinished event. The event fires only when its verifier says Y. Then

$$c_t = \mathbb{I}[k_t \leq K \wedge v_t = Y], \quad (2)$$

$$k_{t+1} = k_t + c_t, \quad r_t^{\text{sem}} = \frac{\lambda}{K} c_t. \quad (3)$$

A N or U verdict leaves  $k_t$  unchanged. The tracker prevents repeated reward for the same landmark, enforces event order, and guarantees  $\sum_t r_t^{\text{sem}} \leq \lambda$ . Because only the pending event is checked, at most one event fires per turn even if the agent performs several

actions. This avoids ambiguous credit when a model emits an action batch but can delay recognition of adjacent events. The tracker retains a confirmed event after later image changes, so a completed event cannot be unconfirmed by viewpoint drift.

The base destination term is the weighted-success reward already used by ActiveVLN, with nDTW weight zero. We add a fixed success floor  $b$  to both arms:

$$R(\tau) = \sum_t r_t^{\text{sem}} + S(\tau) \left[ b + \alpha \left( 1 - \min \left( \frac{d_T}{3}, 1 \right) \right) \right], \quad (4)$$

where  $S(\tau)$  is simulator success after STOP,  $d_T$  is terminal distance in meters,  $\alpha = 15$ ,  $b = 2$ , and  $\lambda \in \{0, 1\}$ . Hence any success receives at least 2 while a failure receives at most 1 semantic point. This priority holds for this nonnegative reward configuration; it does not imply that the verifier is correct. The bound also makes failure reward interpretable: partial event completion can distinguish two unsuccessful attempts without allowing a failed route to outrank a successful one solely through event bonuses. The STOP event is distinct from simulator success: it demands visual proximity to its instruction target, while  $S(\tau)$  uses the simulator’s goal region. Disagreement between these signals is precisely a case the proposed reward aims to expose.

For each training episode, we sample two trajectories and normalize their returns within the group for GRPO [11]. Event rewards are emitted after each turn, but the current trainer sums them before constructing a trajectory advantage. Consequently, the implementation changes which trajectories are preferred, not which individual turn receives the policy gradient. Turn-level event timing is available in the log for future credit-assignment work.

### 3.3. Verifier failure policy

The verifier is a frozen Qwen3-VL-8B-Instruct model. A service timeout, malformed response, or uncertain decision yields zero event reward; it does not end the episode. The reward requires no reference action prefix. Simulator pose is used only for reward verification, never added to the actor observation. In the pilot, the fixed event manifest is matched to both episode ID and exact instruction text to avoid assigning another episode’s event list; the main runs use online parsing. The verifier sees two RGB frames resized to fit within  $448 \times 448$  pixels, the event type and target, its source phrase, the executed action text, translation magnitude, and vertical displacement. Its output is a deterministic one-letter Y/N/U response; generated explanations are not treated as confidence scores. This makes the reward externally auditable through a saved per-turn verdict, although a forced-choice label is still

fallible. The HTTP verifier is separated from the simulator so it can be frozen and logged independently of policy updates.

## 4. Experimental Setup

Environment and policy. We use R2R train in VLN-CE, rendered from Matterport3D scans [3], the ActiveVLN implementation, and an SFT-initialized Qwen2.5-VL-3B policy. Matterport3D data and derived imagery are subject to its [academic terms of use](#). Each turn contains at most three grouped action commands. The configured limits are 12 turns and 36 commands; the strict budget check can permit one extra command, and a command may execute several simulator steps. Actor sampling uses temperature 1.2 and two independently sampled rollouts per item. GRPO trains on two A800 GPUs at learning rate  $10^{-6}$  with batches of four episodes. The destination-only control sets  $\lambda = 0$ ; the event arm sets  $\lambda = 1$ . Both arms have the same SFT initialization, destination reward, data order within seed, and optimizer budget.

Pilot and corrected main training. We first replay eight training episodes with expert actions solely to inspect expected events, then run a two-step reward-wiring pilot with curated events. These replays are neither held-out labels nor training results for the main comparison. For the main study, each arm runs 64 optimization steps at seeds 11, 22, and 33: 256 train-episode exposures and 512 sampled rollouts per arm and seed. The event arm uses automatically parsed events. A preflight check detected that a fixed per-request sampling seed duplicated every two-rollout GRPO group and yielded zero actor gradients; we excluded that attempt, removed the per-request seed while retaining the engine seed, and reran all six models. Every corrected group has distinct sampled actions. The archive records the excluded attempt and corrected patch.

Full val-unseen evaluation. SR is the fraction of trajectories satisfying the simulator goal condition; SPL also accounts for path efficiency. Mean terminal goal distance is in meters. Training rollout success and return are diagnostics, not held-out outcomes; returns differ in definition between arms. The SFT model and six corrected step-64 checkpoints run on the identical manifest of all 1,839 R2R val-unseen episodes from 11 scenes. We use one generation per episode, temperature 0.2, top- $p$  0.8, a 12-turn limit, and at most 76,800 image pixels. Four Habitat shards per checkpoint divide the manifest without overlap. The evaluator uses

Table 2. Observed pilot results on eight training episodes. Each arm ran two optimizer steps and produced 16 sampled rollouts. The event arm’s larger mean return is expected from the added reward and does not indicate better navigation.

| Metric                   | Destination only | EventTrace |
|--------------------------|------------------|------------|
| Successful rollouts      | 13/16            | 13/16      |
| Mean trajectory return   | 9.325            | 10.455     |
| Mean event progress      | –                | 0.656      |
| Confirmed event hits     | –                | 23         |
| Uncertain verifier turns | –                | 33         |
| Verifier service errors  | –                | 0          |

the same prompt and environment-specific OpenAI-compatible request and image-resize shim for all seven models. The completion gate checks the exact episode-ID set and zero inference errors. Episode IDs are paired within each seed, but stochastic decoding uses only one generation per checkpoint and is an additional source of variation.

Blind verifier review. Separately, we collect shortest-path follower trajectories from 16 different val-unseen episodes, of which 12 yield at least one automatically parsed event. Before querying the verifier, one AI-assisted reviewer inspects RGB image pairs, executed actions, and motion and fixes 49 three-way completion labels across those 12 episodes. For the reported replay, both frames use the training client’s 448-pixel JPEG encoding. The sample includes apparent transitions, controls, and final stops. The original AI review was independent of verifier predictions but had no human ground truth. Subsequently, two independent humans labeled the same answer-free, randomized 49-item package without access to predictions or prior labels. Their files were locked before comparison. We report their agreement separately and do not treat unresolved labels as verifier ground truth. Selected class proportions are not representative of deployment traffic.

## 5. Results and Analysis

Reward reaches optimization. In the first optimizer step, both arms sample the same eight action sequences and attain seven successes. For every matched trajectory, the difference in total return equals its semantic event score, including a failed episode 7110 trajectory that receives 0.5 for completing “descend stairs” but never reaches the destination. The modified return is consumed by the training reward manager, and both arms save step-2 checkpoints after nonzero policy updates. This verifies the wiring of Eq. 4; it is not evi-

dence that event reward improves the policy.

What the verifier recognizes. In diagnostic expert replays, four of eight episodes receive full event credit, two receive partial credit, and two receive none; the tracker confirms 11 of 17 curated events. For episode 2889, seeing the sink from outside the bathroom produces uncertain verdicts before entrance, while later motion and view changes confirm enter. A rotation followed by premature STOP yields zero semantic credit in a negative replay. There are also false negatives: one shallow stair descent is rejected by the vertical-motion threshold, and one landmark is never confirmed. Because the events were curated using these episodes and the replays lack independent annotations, 11/17 must not be reported as verifier recall or accuracy.

Main training diagnostics. All six corrected runs finish 64 steps, with 256 diverse GRPO groups per arm and seed. Training-rollout success is 111/512 versus 119/512 (control versus event) for seed 11, 108/512 versus 119/512 for seed 22, and 88/512 versus 82/512 for seed 33. These on-policy samples change as training proceeds, so they are not comparable to the held-out evaluation. The event verifier records 272, 275, and 243 confirmed event hits over the three event-arm runs. These counts are not event accuracy: events are automatically parsed and no independent labels cover these rollouts.

Complete val-unseen results. Table 3 gives all seven checkpoints on the same 1,839 episodes. Within matched seeds, EventTrace changes SR by +0.65, +0.44, and –2.88 percentage points, and SPL by +0.58, +0.49, and –3.49 points. The mean paired seed change is  $-0.60 \pm 1.98$  points for SR and  $-0.80 \pm 2.32$  points for SPL (mean  $\pm$  sample standard deviation over three seeds). At the episode level, event-only versus control-only successes are 157 versus 145, 197 versus 189, and 155 versus 208. This variation does not support a consistent improvement. All seven evaluations have exact 1,839-episode coverage and zero logged inference errors. The earlier 16-episode, two-step probe yielded 6/16 successes for every arm; it was exploratory and is superseded by the complete comparison.

Verifier failure analysis. Table 4 compares the frozen verifier with labels fixed before its predictions were obtained. Seven of eight reviewed completions receive Y, but only 7 of 15 Y predictions match a reviewed completion; four are on clear non-completions and four on ambiguous turns. One false positive occurs after the

Table 3. Complete R2R val-unseen evaluation. Every row uses the same 1,839 episodes from 11 scenes, one generation per episode, and zero inference errors. Goal distance is in meters. C: destination-only control; E: EventTrace.

| Checkpoint | Successes | SR (%) | SPL (%) | Dist. |
|------------|-----------|--------|---------|-------|
| SFT        | 544       | 29.58  | 28.69   | 7.03  |
| Seed 11 C  | 479       | 26.05  | 25.32   | 7.75  |
| Seed 11 E  | 491       | 26.70  | 25.91   | 7.63  |
| Seed 22 C  | 566       | 30.78  | 30.14   | 6.91  |
| Seed 22 E  | 574       | 31.21  | 30.63   | 6.98  |
| Seed 33 C  | 502       | 27.30  | 26.65   | 7.14  |
| Seed 33 E  | 449       | 24.42  | 23.16   | 7.76  |

Table 4. Training-encoded replay of 49 selected val-unseen event transitions from 12 episodes. Rows are blind review labels and columns are frozen verifier predictions. U means evidence insufficient; the single-AI review has no independent human adjudication.

|          | Pred. Y | Pred. N | Pred. U |
|----------|---------|---------|---------|
| Review Y | 7       | 0       | 1       |
| Review N | 4       | 13      | 12      |
| Review U | 4       | 1       | 7       |

agent has already entered the requested room, exposing a failure to distinguish current state from an event newly completed this turn. Motion gates alone do not eliminate this error. These selected examples cannot establish deployment precision, but they make verifier reliability a concrete limitation of the present reward.

Independent human review. The two blinded human reviewers agree on 33/49 three-way labels (67.35%; Cohen’s  $\kappa = 0.459$ ). Agreement is 17/23 for enter, 6/10 for pass, and 10/16 for stop\_near; 16 cases remain disputed. The brief rationales include interpretations of motion toward a landmark as completion and cases where a previously visited landmark cannot be established from the supplied before/after views. This shows that the local evidence and event rubric can be under-determined; it does not isolate model error from label ambiguity. No model-versus-human precision or recall is computed before independent adjudication, and this deliberately selected pilot cannot estimate population accuracy.

## 6. Discussion and Limitations

The main risk is reward misspecification: the VLM may confuse doorways, and displacement gates may miss small valid movements. One-at-a-time verification may miss events completed together. Before/after frames limit timing to a turn; trajectory-level GRPO

offers limited credit assignment to the decisive action.

The complete val-unseen comparison shows a negative mean effect and large seed variation, but 64 steps expose only 256 training-episode instances per run. For EventTrace, a larger training budget and repeated decoding have not been evaluated. The selected 49-example review now has two independent human labelers, but their disagreements have not been adjudicated; instruction parsing is also unaudited. We cannot isolate noisy verification, parser errors, reward scale, budget, or optimization as the cause. Separately sampled and adjudicated verifier and parser sets, followed by ablations of ordering, abstention, gates, reward budget, and turn-level advantage, are needed. This audit cannot establish event precision or recall.

Turn-level credit assignment pilot. We tested a group-four, actor-only return-to-go/leave-one-out estimator inspired by RLOO [1] and REINFORCE++ [4]. Each executed turn receives its remaining return relative to active peer rollouts; observation tokens are masked. The candidate also adds training-only simulator geodesic progress, so both credit assignment and reward change. From the same SFT state and 256 train rows, it and a destination-only GRPO control ran 64 steps; all candidate actor gradients were nonzero. On a frozen 256-episode val-seen manifest (53 scenes, exact coverage, zero inference errors), candidate/control successes were 68/78: paired SR  $-3.91$  and SPL  $-3.72$  points. The prespecified joint  $+2$ -point gate failed, so this mechanism was not expanded. One seed and decode cannot isolate turn-level credit assignment or establish unseen-scene gains. This is not a deployable semantic reward.

Credit-reward factorial follow-up. To separate the two changes in that pilot, we completed a four-arm, group-four comparison from the same SFT initialization and seed, with 256 training rows and 64 optimizer steps per arm. The arms cross trajectory-level GRPO versus turn-level leave-one-out updates with terminal-only versus additional training-time geodesic progress. On a new, disjoint 256-episode val-seen screen (50 scenes; complete coverage and zero inference errors), the outcome-only GRPO control succeeds on 97 episodes. The combined turn/progress, terminal-only turn, and progress-augmented GRPO arms succeed on 96, 98, and 106, respectively. Their paired SR/SPL changes against control are  $(-0.39, -0.14)$ ,  $(+0.39, +0.68)$ , and  $(+3.52, +1.51)$  percentage points. Although progress-augmented GRPO shows a positive SR difference, its SPL difference misses the frozen joint  $+2$ -point advancement gate. Descriptive scene-cluster

407 95% intervals for that arm span zero:  $[-2.24, 9.26]$  SR  
408 and  $[-4.21, 7.17]$  SPL points. An independent per-  
409 episode recount agrees with the paired analysis after  
410 a documented null-ID schema repair to the verifier;  
411 the reward, evaluator, and metric formulas were un-  
412 changed. These one-seed development results do not  
413 establish unseen-scene benefit, and the turn-level and  
414 trajectory-level rules also differ in effective token up-  
415 date scale. The process signal uses privileged geodesic  
416 distance, not the semantic event verifier.

417 Token-scale-matched terminal credit. The factorial's  
418 turn-level estimator divided each turn contrast by its  
419 action-token count, whereas the trajectory GRPO con-  
420 trol applied an order-one advantage to each action to-  
421 ken. We therefore tested a new group-four terminal-  
422 outcome-only leave-one-out estimator that standard-  
423 izes the active-peer contrast and broadcasts it to each  
424 action token in the turn. From the same SFT ini-  
425 tialization, it ran 64 steps on 256 train rows. On a  
426 third, disjoint 256-episode val-seen development screen  
427 (38 scenes), candidate/control successes are 108/103;  
428 paired SR and SPL are +1.95 and +1.26 percent-  
429 age points. Each arm covers every episode once with  
430 no inference errors, and an independent raw recount  
431 agrees. Exploratory scene-cluster intervals span zero:  
432  $[-4.50, 8.04]$  SR and  $[-5.11, 7.26]$  SPL points. Both  
433 improvements fall short of the predeclared joint +2-  
434 point gate, so the original conditional three-seed ex-  
435 pansion was skipped. A subsequent user-directed, ex-  
436 plicitly post-result group-eight sensitivity and group-  
437 four three-seed expansion were conducted separately.  
438 This one-seed screen does not establish unseen-scene  
439 benefit or validate a semantic reward.

440 Post-result group-size sensitivity. Keeping the 64-  
441 step, 256-row prompts and terminal-RLOO rule fixed,  
442 we sampled eight rather than four trajectories per in-  
443 struction. This doubles simulator rollout work. The  
444 group-eight run had logged nonzero actor gradients on  
445 63 of 64 steps; the remaining step had an all-failure  
446 group and zero advantage. On the same 256-episode  
447 val-seen screen, the group-eight checkpoint succeeds on  
448 112 episodes, versus 103 for the destination-only con-  
449 trol and 108 for the group-four checkpoint. Its paired  
450 SR/SPL differences are +3.52/+2.82 points versus con-  
451 trol and +1.56/+1.56 points versus group four. Every  
452 arm covers all 256 episodes across 38 scenes, with no  
453 inference errors, and an independent raw-stat recount  
454 agrees with the compact episode records. Descriptive  
455 scene-cluster intervals for group eight versus control  
456 span zero:  $[-4.92, 12.13]$  SR and  $[-5.28, 11.10]$  SPL  
457 points. This comparison was requested after the group-

four advancement gate failed; the group-eight result  
does not retroactively satisfy that gate. It is one-seed,  
adaptively selected development evidence, with group  
size confounded by rollout compute. The separately  
authorized group-four three-seed expansion is still run-  
ning and has no result in this manuscript version.

Multimodal GAE actor-critic pilot. We also tested  
masked token-level GAE with a separately initial-  
ized multimodal value head and terminal-outcome-only  
feedback. The group-four actor and critic started from  
the same SFT model; 64 optimizer steps on 256 train  
rows had finite value losses and nonzero actor and critic  
gradients at every step. On an adaptively reused 778-  
episode val-seen development screen (53 scenes), the  
GAE candidate succeeds on 22 episodes versus 272  
for the same-seed destination-only control: paired SR  
 $-32.13$  and SPL  $-31.56$  percentage points. Both arms  
have exact episode coverage and zero inference errors;  
two independent raw-stat recounts and a compact ex-  
port recount agree. Descriptive scene-cluster inter-  
vals are  $[-36.83, -27.43]$  SR and  $[-36.19, -26.95]$  SPL  
points. Mean path lengths are 2.79 versus 6.53 m, con-  
sistent with shorter candidate trajectories but insuf-  
ficient to identify why learning failed. The prespeci-  
fied joint +2-point scale gate failed, so the conditional  
three-seed GAE expansion was skipped. This one-seed  
development result is not evidence about unseen-scene  
generalization.

Exploratory policy-prefix follow-up. Separately from  
EventTrace, we tested whether changing the training-  
state distribution helps destination-only GRPO. For  
each of three seeds, a branching curriculum replayed  
a training-only policy prefix before sampling two con-  
tinuations; its matched control started at the task ori-  
gin. Both arms used the same 512 train episodes, SFT  
initialization, destination reward, and 128 optimizer  
steps. All six checkpoints covered the complete 1,839-  
episode val-unseen split with zero inference errors. Ta-  
ble 5 shows sharply different seed effects and a negative  
three-seed mean. Exploratory scene-and-seed 95% in-  
tervals for the mean differences are  $[-6.51, 4.02]$  SR and  
 $[-6.16, 4.85]$  SPL points. On the 1,583 episodes outside  
the fixed 256-episode development screen, the mean  
changes remain  $-1.22$  SR and  $-0.55$  SPL points. Sepa-  
rate stochastic decodes of the overlapping 256 episodes  
changed success outcomes for 20–41 episodes per check-  
point, so the small screen is an unstable model-selection  
signal. This follow-up changes start state and remain-  
ing budget together; it does not identify why Event-  
Trace failed or establish a benefit from branching. The  
checked paired episode records and training audits ac-

Table 5. Exploratory 128-step policy-prefix follow-up on all 1,839 val-unseen episodes. Differences are branching minus a same-data, from-scratch destination-only control, in percentage points. Mean  $\pm$  sample standard deviation is over three training seeds; each model has one stochastic decode per episode.

| Seed          | $\Delta$ SR      | $\Delta$ SPL     |
|---------------|------------------|------------------|
| 11            | -7.18            | -6.86            |
| 22            | +0.49            | +0.80            |
| 33            | +3.26            | +4.41            |
| Mean $\pm$ SD | -1.14 $\pm$ 5.41 | -0.55 $\pm$ 5.75 |

Table 6. Exploratory 128-step, destination-only group-size follow-up on all 1,839 val-unseen episodes. Differences are four samples minus a matched two-sample control, in percentage points; the four-sample arm uses twice the simulator rollouts.

| Seed          | $\Delta$ SR      | $\Delta$ SPL     |
|---------------|------------------|------------------|
| 11            | +0.65            | +0.96            |
| 22            | +1.36            | +1.44            |
| 33            | +4.30            | +5.12            |
| Mean $\pm$ SD | +2.10 $\pm$ 1.93 | +2.51 $\pm$ 2.27 |

company the manuscript.

Group-four optimizer follow-up. A separate destination-only comparison tests four sampled trajectories per GRPO group against the same-data, two-sample control. Both arms use the same SFT initialization, 512 unique training episodes, and 128 optimizer steps at each of three matched seeds; the four-sample arm uses twice as many simulator rollouts. All six checkpoints cover the identical 1,839-episode val-unseen manifest with zero inference errors. Table 6 shows positive paired differences in each seed: the mean is  $+2.10 \pm 1.93$  SR and  $+2.51 \pm 2.27$  SPL percentage points (mean  $\pm$  sample standard deviation over seeds). On the 1,583 episodes outside the fixed 256-episode screen, the means are  $+1.94$  SR and  $+2.40$  SPL points. The largest seed-33 gain is  $+4.30$  SR points, while exploratory 11-scene bootstrap intervals for seeds 11 and 22 cross zero. One stochastic decode per checkpoint and only three training seeds limit the inference. This comparison does not identify a better semantic reward; subsequent reward comparisons must hold group size at four in both arms. The complete paired episode records, training audits, and independent recomputation script accompany the paper.

Instruction-grounded reward representation. We also probe a potential reward that combines a causal

progress encoder from frozen navigation-policy states with a SigLIP image-instruction encoder adapted on training-scene goal panoramas and group-four successful/failed routes. SigLIP text is padded to a fixed 64 tokens, and offline scoring uses the same one-image request shape as the online service. Each component is normalized by its fit-scene mean absolute margin; a fixed 1:1 mixture is applied without development-set tuning. On 52 development pairs, successful-route ranking rises from 37/52 for the temporal term and 32/52 for the visual term to 43/52 for the mixture; correct-versus-wrong-instruction preference is 39/52. On 38 new episode-disjoint, train-scene pairs, the mixture scores 30/38 route ranks and 34/38 instruction comparisons. A previously inspected 48-pair scene audit scores 41/48 and 37/48. These selected offline pairs share train scenes or prior audit use, so they provide a screen for an online group-four reward comparison, not evidence of val-unseen navigation gain.

Online fused-reward screen. We added this frozen potential as a bounded terminal bonus to the destination reward and trained a matched group-four policy for 64 steps at seed 11. Its training audit verifies 1,024 reward requests for 1,024 rollouts across 256 matched train episodes. On the fixed 256-episode val-unseen screen, the fused-reward checkpoint achieves 68/256 successes versus 80/256 for the same-data, same-group-size destination-only control: paired SR changes by  $-4.69$  percentage points and SPL by  $-4.34$  points. Both arms cover exactly 256 episodes in 11 scenes with zero inference errors. This single-seed screen does not support scaling this particular reward, despite its offline ranking results. A failure-aware encoder trained on unsuccessful pairs improves a separate 106-pair train-scene ranking audit from 69/106 to 79/106, but direct fusion with the image-instruction score weakens reused instruction-matching probes. Those representation findings do not establish a navigation improvement.

Failure-only reward screen. We next applied the new temporal potential only to unsuccessful rollouts, leaving successful rollouts on the original destination reward. A matched seed-11 group-four policy completed 64 steps on the same training episodes; its audit found 854 reward requests for 854 unsuccessful rollouts and no extra bonus for 170 successful rollouts. On the same fixed 256 val-unseen episodes, it achieves 63/256 successes versus the destination-only control’s 80/256: paired SR changes by  $-6.64$  points and SPL by  $-6.05$  points. Both models cover 11 scenes without inference errors. The exploratory scene-cluster 95% intervals

are  $[-11.07, -2.29]$  SR and  $[-10.66, -1.47]$  SPL points. This screen fails the predeclared positive SR-and-SPL scale gate; we therefore do not run a three-seed scale-up for this reward. Training-only failed-pair ranking (622/970 in all-failure groups) does not transfer into a navigation benefit. A post-hoc behavior check finds maximum-turn termination on 187 candidate episodes versus 107 control episodes, while mean traveled path length falls from 6.74 to 5.71 m; these differences motivate a stopping-aware reward test but do not identify the cause of the regression. An offline equal mixture with the adapted visual score also fails its train-scene development gate (68/125 failed-pair ranks versus 90/125 for the temporal term), so its audit remains unopened.

Stop-aware reward screen. To test whether terminal-mode conditioning mitigates this failure, we gave the frozen temporal bonus only to unsuccessful voluntary STOP trajectories. Successes retained the destination reward, while turn-limit and format failures received no bonus. The matched seed-11 group-four policy completed 64 steps on the same training episodes. Its audit found exactly 328 scorer requests for 328 unsuccessful voluntary stops across 1,024 rollouts, with nonzero actor gradients at every step. On a new, disjoint 256-episode val-unseen screen covering 10 scenes, the candidate succeeds on 45/256 episodes versus 93/256 for its same-seed destination-only control. Paired SR and SPL changes are  $-18.75$  and  $-17.43$  percentage points; both arms have zero inference errors. The predeclared positive SR-and-SPL gate failed, so this reward received no three-seed expansion. Post-hoc analysis finds mean path length 4.44 versus 6.93 m and only three candidate episodes with a non-null termination reason versus 116 controls, consistent with frequent early stops. On the candidate's train rollouts, its frozen raw score has AUC 0.486 for distinguishing 18 unsuccessful STOP endpoints within 3.5 m of the goal from 310 farther endpoints; only 11 of 183 positive-bonus stops are within 3.5 m. This train-scene diagnostic does not establish causality, but it shows that failure-pair ranking is insufficient calibration for rewarding STOP.

Mode-stratified ordinal screen. To remove a direct reward offset between unsuccessful STOP and turn-limit trajectories, we next ranked frozen temporal scores only within the same terminal mode of each four-rollout all-failure group; any-success groups retained the original outcome reward. This is a trajectory-level group comparison, not true turn-wise credit assignment: the current GRPO implementation broadcasts a group-normalized total reward to all action

tokens. The 64-step seed-11 training audit verified 1,024 rollouts, 149 all-failure groups with active ordinal rewards, 827 scorer requests for exactly 827 failures, and a nonzero actor update at every step. On a third fixed, nonoverlapping 256-episode val-unseen screen spanning nine scenes, the mode-stratified candidate succeeds on 59/256 episodes versus 73/256 for its same-data, same-seed group-four outcome-only control. Paired SR and SPL change by  $-5.47$  and  $-5.23$  percentage points, with zero inference errors in either arm. The exploratory nine-scene bootstrap 95% intervals are  $[-9.06, -1.95]$  SR and  $[-8.98, -1.74]$  SPL points. Among failures, the candidate ends with an unforced model STOP 148 times versus 84 for the control; this post-hoc association does not identify a causal mechanism. The predeclared positive SR-and-SPL scale gate failed, so no 128-step or three-seed extension was run for this reward. Absolute success rates across the three 256-episode screens are not directly comparable because their episode sets differ.

Privileged turn-wise credit upper bound. The preceding trajectory-level rewards broadcast one return to every action token. To isolate whether finer credit assignment can help, we tested a training-only oracle that labels each executed turn by its change in simulator geodesic distance. In all-failure groups of four, it centers return-to-go among active rollouts at each turn and gives no auxiliary advantage to generated STOP tokens; groups containing a success retain destination-outcome GRPO. Geodesic distance is never a policy observation, but this oracle is not a learned semantic reward. The same-data seed-11 pilot completed 64 steps. An independent training audit checked 256 four-rollout groups, 11,450 turns, active contrast in all 141 all-failure groups, and nonzero actor gradients on all 64 steps. On a fourth, pre-frozen 256-episode val-unseen screen spanning nine scenes, it succeeds on 81 episodes versus 72 for its outcome-only control: paired SR  $+3.52$  and SPL  $+3.76$  percentage points, with zero inference errors. The exploratory scene bootstrap 95% intervals span  $[-2.73, 9.13]$  SR and  $[-2.58, 9.28]$  SPL points. This one-seed screen selected a same-checkpoint full val-unseen recheck. Across all 1,839 episodes, the oracle succeeds on 602 versus 545 for the control: paired SR  $+3.10$  and SPL  $+3.05$  points. On the 1,583 episodes outside the reused screen, it succeeds on 521 versus 481: SR  $+2.53$  and SPL  $+2.50$  points. Both evaluations have exact coverage and zero inference errors. This remains a post-screen, one-seed development result using privileged training labels; it neither validates EventTrace's semantic verifier nor establishes a deployable learned reward.

Table 7. Matched group-four, 512-row, 128-step privileged turn-wise oracle versus destination-only control on all 1,839 val-unseen episodes. Differences are oracle minus control in percentage points.

| Seed          | SR counts O/C | $\Delta$ SR      | $\Delta$ SPL     |
|---------------|---------------|------------------|------------------|
| 11            | 513/450       | +3.43            | +3.45            |
| 22            | 525/545       | -1.09            | -0.61            |
| 33            | 585/492       | +5.06            | +4.70            |
| Mean $\pm$ SD | —             | $+2.47 \pm 3.18$ | $+2.51 \pm 2.78$ |

We next scaled this privileged mechanism to 512 distinct training episodes and 128 optimizer steps at each of three seeds, with four rollouts per group in both oracle and destination-only arms. All six checkpoints were decoded once on the same 1,839 val-unseen episodes, with exact coverage and zero inference errors. Table 7 gives paired results. The mean oracle change is  $+2.47 \pm 3.18$  SR and  $+2.51 \pm 2.78$  SPL points (sample SD over seeds); seed 22 instead loses 1.09 SR points. On the 1,583 episodes outside the reused 256-item development screen, the means are  $+2.63$  SR and  $+2.64$  SPL points. Exploratory scene-and-seed intervals for the full-set mean are  $[-1.49, 6.05]$  SR and  $[-1.20, 5.78]$  SPL points. Thus the observed mean is positive, but the seed variation and intervals do not support a consistent gain. The turn reward still uses simulator distance.

An adaptive stopping-boundary pilot tested whether capping geodesic progress credit inside the 3 m success region and penalizing further movement there by 0.1 improves this oracle. Its seed-11 matched  $n=4$  64-step audit verified 256 identical training rows, nonzero actor gradients at every step, and 902 inside-boundary penalty turns. On the reused 256-episode development screen, the candidate succeeds 67 times versus 72 for the unchanged control: paired SR  $-1.95$  and SPL  $-1.62$  points, with zero inference errors. This fails the fixed positive SR-and-SPL scale gate. Neither privileged variant establishes a deployable semantic reward.

Two observation-only progress representations were screened before online use. An instruction-conditioned scalar potential fit to cached policy RGB histories reached only 48.88% balanced forward/regression accuracy on its small train-scene development split. A joint two-view antisymmetric LoRA reached 53.52% balanced accuracy and 52.08% correct-versus-wrong-instruction preference at its selected checkpoint; at 9.88% stationary false positives, its forward recall was zero. Neither passed the fixed development gate or entered RL. A further start-anchored, action-conditioned three-view LoRA also failed its small development gate (55.51% balanced direction, 22.22% regression accu-

racy), so it received no audit or online reward test. This leaves distillation of the privileged credit signal into an observation-only representation as open work.

We also screened arrival evidence on a scene-disjoint R2R-train split of audited group-four policy rollouts. A single-frame occupancy head reached AUC 0.6053, but recalled only 8.33% of inside-region views and 5.04% of inside-region views on unsuccessful routes at a pooled false-positive rate below 5%. A separate joint two-view, instruction-conditioned transition head was trained for a fixed 1,024 microsteps on real crossings, far nonarrivals, real near-goal retreats, and identical-view wrong-instruction pairs. On all 758 development pairs from 351 records, its selected checkpoint reached AUC 0.6430, versus 0.5177 for a text-only control and 0.5211 after same-scene image-pair shuffling. For identical image pairs with different instructions, the correct instruction ranked higher in only 98/176 cases (55.68%). A CPU-only source audit confirms that these pairs share the exact simulator states; their wrong-instruction labels are geometric alternate-goal proxies, without independent semantic adjudication. Weak grounding and label ambiguity cannot be separated by this diagnostic. Nevertheless, at 4.89% pooled false positives it recalled only 15.63% of crossings and 13.45% of unsuccessful crossings, below the frozen 55% and 50% gates. These exploratory screens reused development scenes across method design; neither opened its reserved train-scene audit or entered online RL. Better AUC therefore supplies no evidence of a navigation gain from this learned reward.

## 7. Conclusion

We formulated a path-free semantic process reward for closed-loop VLN that verifies ordered instruction events against executed motion and before/after views. A wiring pilot confirms that the bounded reward enters online GRPO. In three matched 64-step training runs and complete 1,839-episode val-unseen evaluation, EventTrace has a mean paired SR change of  $-0.60$  percentage points relative to destination-only training, with inconsistent seed effects. Two independent human reviewers agree on 33/49 selected transitions, but the remaining cases lack adjudication and the selected sample cannot establish verifier accuracy. The current method is reproducible; reliable semantic verification and a navigation benefit remain unproven.

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